ZACH PARK

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
Master of Science in Mechanical Engineering - Research *May 2028*
Research: Safe AI Lab (Prof. Ding Zhao) - Safe Robot Learning, Multi-Hazard Constrained RL, Multimodal Robot Safety
Current Coursework: Machine Learning and AI for Engineers, Trustworthy AI

The Cooper Union for the Advancement of Science and Art New York, NY
Bachelor of Engineering in Mechanical Engineering, Minor in Bioengineering, GPA: 3.86/4.0 (Summa Cum Laude) *May 2026*
Related Coursework: Manufacturing Engineering, Design & Prototyping, Mechatronics, Signal Processing, Materials Science

PROFESSIONAL EXPERIENCE

Product Design Intern, HUROTICS Inc. *Seoul, South Korea | Jun. 2026 – Aug. 2026*

- Designed company's first gait test robot in SolidWorks and developed LabVIEW-based actuator control architecture for automated exosuit validation, resulting in company-funded $10K production build
- Improved gait-labeling accuracy by 13% through ToF-IMU sensing system with ML-based classification, sensor hardware, Teensy firmware, and MATLAB signal processing
- Detected upcoming terrain 34% earlier than previous system by developing Python LiDAR-camera-IMU perception pipeline using multimodal sensor fusion, optical-flow tracking, and regression-based prediction
- Collaborated with apparel and embedded teams to design self-locking anchors, actuator docking, and thermal-management hardware for next-generation exosuit, with 2 components selected for final production design

Mechanical Design Intern, Mount Sinai Hospital BioDesign *New York, NY | Aug. 2025 – May 2026*

- Reduced fabrication time by 75% and material cost by 40% by prototyping reusable silicone intracerebral hemorrhage (ICH) brain-and-skull phantom in Fusion 360 for neurosurgical training and device validation
- Created patient-specific chronic subdural hematoma (CSDH) testbed from CT anatomy with under 1 mm cranial deviation, replaceable dura, silicone brain, and configurable burr-hole access for medical device testing
- Increased endoscopic field of view by 17% through 4 CAD design iterations with neurosurgeons and industry partners

Summer Research Intern, Wake Forest University Winston-Salem, NC | May 2025 – Aug. 2025

- Segmented 19 spinal muscles from pre/post-flight MRI of 9 long-duration ISS astronauts in Mimics to generate subject-specific inputs for LS-DYNA human-body FE models and scale GHBMC M50 parameters
- Quantified region- and sex-specific muscle morphology, identifying lumbar cross-sectional area up to 2.04× GHBMC M50 reference and establishing scaling factors for subject-specific biomechanical simulations

PROJECT EXPERIENCE

Active Armrest Robot for Microsurgery Clinicians, Cooper Union *New York, NY | Aug. 2025 – May 2026*

- Led 3-engineer interdisciplinary team with Mount Sinai clinicians to develop stool-mounted active armrest integrating C++ control, lead-screw actuation, ToF sensing, and machined hardware for hands-free height adjustment
- Developed and CNC-machined 6061-aluminum two-link arm with GD&T and tolerance-controlled fits, validating 7 kg design load through FE with factor of safety >7
- Lowered trapezius EMG RMS by 30%, drawing deviation by 23%, and forward-head offset by 11% through sEMG signal processing, DWPose computer-vision posture tracking, and precision-task validation

Wearable Sensor for Dialysis Access Monitoring, Cooper Union *New York, NY | Jan. 2025 – May 2026*

- Partnered with Mount Sinai nephrologists to engineer dual-axis piezoelectric wearable for arteriovenous fistula (AVF) vibration monitoring using orthogonal sensing, differential amplification, and LabVIEW-based signal processing
- Fabricated <$200 vascular test replacing $3,000+ commercial pump, supporting 30+ validation trials and contributing to 2nd Place and $15K award in 2025 NIH DEBUT Challenge

Deep Learning for Electromagnetic Forming, Technische Universität Dresden *Dresden, Germany | May 2024 – Aug. 2024*

- Integrated LS-DYNA electromagnetic-forming simulations with Python/PyTorch deep-learning pipeline, training Temporal Fusion Transformer to predict sheet-metal forming depth from material, process, and current data
- Cut predictive coefficient of variation by up to 66.7% by guiding uncertainty-driven DoE on 210-sample dataset, using parameter-specific Monte Carlo dropout to prioritize 29 additional physical tests with Fraunhofer IWU

SKILLS

Languages: Python, C, C++, MATLAB
Tools: SolidWorks, Fusion 360, ANSYS, LS-DYNA, LabVIEW/NI-DAQ, ROS 2, PyTorch, TensorFlow, scikit-learn, OpenCV
Methods: Rapid Prototyping, GD&T, Tolerance Analysis, DFM/DFA, FEA, Sensor Fusion